

Test 8

Hi-tech fabric

Clothes designers have lots of exciting ideas, but this one is (1) interesting. Imagine if your T-shirt changed colour whenever you put your hand on it, or the moment you stepped into a noisy room! A new fabric has been designed which changes colour when it is (2) to touch or sound. This unusual reaction (3) when ordinary fabric is dyed with special ink, known as thermochromatic ink, which changes colour when the temperature increases or decreases. So when someone's hand comes into (4) with the fabric, its natural heat (5) with the chemicals in the ink (6) in colourful, but temporary fingerprints.

To make the fabric change colour when music is playing, special metal wires are woven into it, which are (7) to a tiny microcomputer. The energy from the sound (8) the wires to heat up, which in (9) change the temperature of the thermochromatic ink. This means that the fabric keeps on changing colour as long as the sound continues.

READING AND USE OF ENGLISH (1 hour 15 minutes)

Part 1

For questions 1–8, read the text below and decide which answer (A, B, C or D) best fits each gap. There is an example at the beginning (0).

Mark your answers **on the separate answer sheet**.

Example:

0 A mainly B particularly C largely D principally

0	A	B	C	D
	☐	☒	☐	☐

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This unusual reaction (2) when ordinary fabric is dyed with special ink, known as thermochromatic ink, which changes colour when the temperature increases or decreases. So, when someone's hand comes into (3) with the fabric, its natural heat (4) with the chemicals in the ink, (5) in colourful, but temporary fingerprints.

To make the fabric change colour when music is playing, special metal wires are woven into it, which are (6) to a tiny microcomputer. The energy from the sound (7) the wires to heat up, which in (8) change the temperature of the thermochromatic ink. This means that the fabric keeps on changing colour as long as the sound continues.

- | | | | | |
|----------|--------------------|---------------------|--------------------|---------------------|
| 1 | A exposed | B given | C presented | D brought |
| 2 | A appears | B reveals | C occurs | D arrives |
| 3 | A contact | B touch | C access | D connection |
| 4 | A affects | B reacts | C adapts | D responds |
| 5 | A finishing | B concluding | C leading | D resulting |
| 6 | A linked | B set | C combined | D put |
| 7 | A leads | B makes | C causes | D brings |
| 8 | A place | B case | C order | D turn |

Part 2

For questions **9–16**, read the text below and think of the word which best fits each gap. Use only **one** word in each gap. There is an example at the beginning **(0)**.

Write your answers **IN CAPITAL LETTERS** on the separate answer sheet.

Example:

0	T	O															
---	---	---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Giant Penguin

Bones belonging **(0)** a giant penguin that existed around 40 million years ago have recently been unearthed by scientists. They form part of the most complete fossil ever uncovered from the Antarctic. The penguin species, *Palaeudyptes klekowskii*, more commonly **(9)** as the 'colossus penguin', would have **(10)** today's biggest living penguins look small. In fact, they were more or **(11)** the same height as the average human, according to researchers.

By examining those bones, researchers established that the penguins would have stood 2 m tall from toe to beak tip, and weighed as **(12)** as 115 kg. In comparison, the tallest and heaviest living penguin, the emperor penguin, stands 1.1 m high and weighs just under 50 kg. **(13)** large has its advantages, as bigger penguins can dive underwater to hunt fish **(14)** significantly longer periods of time compared to smaller species. A penguin **(15)** size of *Palaeudyptes klekowskii* could have stayed underwater for **(16)** to 40 minutes.

Part 3

For questions **17–24**, read the text below. Use the word given in capitals at the end of some of the lines to form a word that fits in the gap **in the same line**. There is an example at the beginning **(0)**.

Write your answers **IN CAPITAL LETTERS** on the separate answer sheet.

Example:

0

N	O	R	M	A	L	L	Y										
---	---	---	---	---	---	---	---	--	--	--	--	--	--	--	--	--	--

Will it rain today?

Like most people, I **(0)** check the weather forecast before I go out. Knowing what the weather is going to be like helps us plan our **(17)** activities effectively.

NORMAL

DAY

Weather forecasting has a long history. In the past, people recorded changes in the weather to identify patterns which would **(18)** them to forecast with some **(19)** They realised, for example, that high, delicate clouds generally announced the **(20)** of good weather.

ABLE

ACCURATE

ARRIVE

Modern forecasting is more sophisticated, and uses a **(21)** of computer models, monitoring and knowledge of trends. Today's forecasts are based on several different **(22)** Rainfall, wind direction, wind speed and air pressure are all recorded. This means that forecasting is becoming increasingly reliable.

COMBINE

MEASURE

Nature will always be **(23)** though, so we can never really be sure what will happen. It's difficult to forecast further in advance than five days with much **(24)** as small changes can have a big impact on the weather.

PREDICT

CERTAIN

Part 4

For questions **25–30**, complete the second sentence so that it has a similar meaning to the first sentence, using the word given. **Do not change the word given.** You must use between **two** and **five** words, including the word given. Here is an example (0).

Example:

- 0 Prizes are given out when the school year finishes.

PLACE

Prize-giving end of each school year.

The gap can be filled by the words 'takes place at the', so you write:

Example:

0	TAKES PLACE AT THE
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Write **only** the missing words **IN CAPITAL LETTERS** on the separate answer sheet.

- 25 They say that the new sports centre is fantastic.

SAID

The new sports centre fantastic.

- 26 I can't think who this pen belongs to.

WHOSE

I have no is.

- 27 Jane's parents wouldn't let her go to the party.

REFUSED

Jane's parents her to go to the party.

- 28** Jim's house isn't close enough for him to walk to school every day.

AWAY

Jim's house is for him to walk to school every day.

- 29** I recommend buying that phone right now, before it sells out.

BETTER

I think you that phone right now, before it sells out.

- 30** I think John's trying to get you to notice him – he's waving at you.

ATTRACT

I think John's trying to , because he's waving at you.

Part 5

You are going to read a magazine article about a teenager who goes on a road trip across the US with his parents and twin brother. For questions 31–36, choose the answer (A, B, C or D) which you think fits best according to the text.

Mark your answers on the separate answer sheet.

Our Family Great American Road Trip

By Jack Morgan

When most people think of a leisurely family vacation, I guess a cross-country adventure through nine western states in the US plus a Canadian province doesn't automatically spring to mind. However, my parents clearly weren't looking to put their feet up as they organized our summer vacation this year. My dad seemed to have taken it upon himself to plan the modern teenager's worst nightmare: the *ultimate family bonding trip*. 'Not enough people do this anymore' was the line he and my mother spouted in the weeks leading up to our vacation as my twin brother and I shot each other horrified glances that screamed, 'Well, there are probably good reasons why!'

Mid-June arrived. We crammed our car full of suitcases, coolers, and a wild assortment of clothing to combat whatever Mother Nature threw at us. Though I was thrilled, I was sure going to miss my privacy for the next two weeks. But I already had a few coping strategies up my sleeve. With barely enough room left for us inside the car, we set off on 'The Family Great American Road Trip'. As we left behind our spacious home in favor of more 'cozy' accommodation for the next five thousand miles, I wasn't the only one in the car wondering what we were getting ourselves into...

line 31 We shot straight west, out through the Oklahoma panhandle. Though still close to home, we were not without our fair share of adventures. My family finally stopped for the night near a town in Oklahoma, under the darkest star-filled skies in the nation, but only after nearly being blown off the highway in a dust storm. We quickly learned the rule of the road: cattle get the right-of-way! The next day, we were up at dawn and moving again. I sensed a pattern forming.

Desert gave way to mountains. The days began to blend together, but each adventure was still distinct. We left our footprints in the Great Sand Dunes, and moved on to Pagosa Springs,

Colorado. The unpleasant smell of sulfur in that otherwise welcoming tourist town gave us a taste of what we were going to experience later in our journey. Clinging to the mountainside at one point as we climbed to our next destination, we learned that what are marked as jeep roads are best left to jeep drivers. The fusion of ancient and modern could be seen everywhere we stopped. Dodging construction traffic just became routine. Disappointment turned to laughter as we posed by the enormous 'CLOSED' signs in front of the Four Corners Monument.

We passed countless wind farms on our way to Yellowstone National Park, where we watched the setting sun cast its shadows over bear habitats close to the roads. 'Bear jams', policed by park volunteers, were simply expected while driving through, when traffic would back up to let one cross. And our own close encounter with one on the road (and an anxious park ranger) left us with some unforgettable memories. As we drove on through the park, life back home faded away, and even the hotel we stayed in that night, with all its inconveniences, couldn't dent our enthusiasm – a creaky, rustic 95-year-old building with no internet or cell phone services. However, in exchange for putting up with these, I got to experience the most wonderful views I've ever seen.

line 71

After watching a rainstorm blow through from a cozy perch in our hotel in Canada, my family and I made our free-fall back south again. My parents had been right. You don't have to escape to some exotic foreign land to discover something remarkable. It was bittersweet to be home together again, but at least I hadn't fought with my brother in our one hundred hours on the road! Three thousand photos chronicling our crazy journey waited to be sifted through and rediscovered. And I knew that for me, the things I'd remember would be the unplanned, the spontaneous.

- 31 Jack suggests in the first paragraph that his parents
- A had waited until their sons grew up to make such a long journey.
 - B wanted to recreate the type of holiday they themselves had once enjoyed.
 - C were aware that the trip would offer them little chance of relaxation.
 - D seemed frustrated by their sons' protests about the proposed trip.
- 32 In the second paragraph, Jack explains that
- A there were problems he had not anticipated.
 - B his family were all equally excited about what lay ahead.
 - C his family were prepared for any kind of weather.
 - D he was looking forward to the challenge of living without home comforts.
- 33 What does Jack mean when he says they were 'not without our fair share of adventures' in line 31?
- A They had already had a good number of adventures.
 - B They each felt they would like to have more adventures.
 - C They had experienced more adventures than they had hoped.
 - D They felt they had been unfairly deprived of any adventures so far.
- 34 In his description of the trip in the fourth paragraph, Jack sounds
- A eager to discover what lay further along the route.
 - B confused about what he had done on each day.
 - C impatient about the number of times they had to stop.
 - D willing to make the best of unexpected setbacks.
- 35 What does 'these' refer to in line 71?
- A memories
 - B inconveniences
 - C wonderful views
 - D phone and internet services
- 36 What does Jack conclude that going on the trip had taught him?
- A how important it is to prepare for chance occurrences
 - B to appreciate the beauty of the natural world
 - C how hard it can be to get on with close family
 - D to value what your own country has to offer

Part 6

You are going to read an article about poetry. Six sentences have been removed from the article. Choose from the sentences **A–G** the one which fits each gap (37–42). There is one extra sentence which you do not need to use.

Mark your answers on the separate answer sheet.

Poetry

Teenager Peter Rayner writes about his new-found love of poetry.

Until about a year ago, if you mentioned the word 'poetry' to me, my instant reaction would always have been the same: 'Boring!' And I'm sure the response would be the same from many people my age. But then recently I've discovered that by rejecting poetry so completely, I've really been missing out.

It's not that I was previously unaware of poetry. I've been lucky enough to grow up in a household where my parents and older sister love literature, and I've always loved reading, especially crime novels – I read one a week, on average. **37** They seemed to have little to say about what I was feeling.

Then one day I came across an anthology of poems for young people. On the first page, the editor had written an introduction in which he explained that poems are not just there to be studied in classrooms. **38** He reckoned that poetry often helps people to find some kind of peace when times get tough. I was really struck by that, and began to think that maybe poetry did have relevance for me after all.

After I'd read the book, I started to see that poetry is all around us, whether we realise it or not. If you don't believe me, just think of song lyrics, rap and football chants. There were so many times I'd been swept away by the lyrics of various rappers and bands. **39**

So there it was – I'd actually been listening to poetry all along! This made me start reading poetry seriously for the first time. **40** That was a great feeling. Poems spoke to me directly whilst still offering food for thought. I even started to write my own terrible raps! Poetry was now definitely cool as far as I was concerned.

41 Did you know, for example, that spoken word performances, combined with music, were the major literary form of the ancient world? They were popular with Greek storytellers who'd tell their stories while a musician accompanied them on the harp. I suppose the modern day equivalents would be poetry performed to a reggae soundtrack, poems about social observations, or poets drawing on influences from the hip hop scene.

Anyway, I started trawling the internet to see if there were any performance poets who were my age. I found myself googling them and it wasn't long before I discovered some teenage poets who were performing alone or with musicians. **42** I've seen some of them live and they're brilliant. I've now been inspired to have a go at writing more serious poetry and maybe one day I'll try performing it too.

- | | |
|--|--|
| <p>A Their stuff has really taken off online and even at music festivals.</p> <p>B Poems, however, just never appealed to me in the same way.</p> <p>C And I could understand it without having to be taught what it meant.</p> | <p>D In spite of this, I began writing them myself.</p> <p>E All of this is nothing new.</p> <p>F Reading them can provide support for us at any time in our lives.</p> <p>G I'd simply never, ever thought of them as poetry.</p> |
|--|--|

Part 7

You are going to read an article in which young people talk about the environment. For questions **43–52**, choose from the people (**A–D**). The people may be chosen more than once.

Mark your answers **on the separate answer sheet**.

Which person

gives an example of what has inspired positive feelings in people about their local area?

43	
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thinks there should be no choice about adopting green measures?

44	
----	--

feels people their age could spread the word effectively about environmental matters?

45	
----	--

has problems persuading young people that green issues are important?

46	
----	--

initially had trouble finding a green organisation to join?

47	
----	--

explains that receiving recognition for something encouraged them to change their habits?

48	
----	--

shows an understanding of why people aren't focused on green issues?

49	
----	--

understands that the previous generation has witnessed the development of current environmental issues?

50	
----	--

recognises that making major changes could prove unaffordable?

51	
----	--

admits not applying what they'd been taught to their daily lives?

52	
----	--

Young people and the environment

A Julia

We're always being bombarded by news about celebrities in the media, but the same can't really be said for the latest on global warming, can it? Maybe if things like that hit the headlines more often, there'd be more involvement. My generation is very creative, finding different ways to interact with people using social media, music and so on. And if that creativity could be put to good use to pass on the eco message, maybe more people would take note. But instead they're delighted that the temperature seems to rise each summer – they sit around in the local park soaking up the sun and don't bother to think about what the cause of this might be – and I do get that. But perhaps if the consequences of not taking care of the environment were more apparent, it wouldn't be so easy to disregard.

B Harald

I'm aware that what my parents have seen gives them a different take on things to me – like climate change, which has been an issue all their lives. And I still haven't fully faced up to the reality of what we're doing to the Earth – I know that's not great. But recently I did take part in a competition about living more sustainably – and I won a prize. That kind of reward for improving my lifestyle motivated me to have a good think about what I could do. And I really do believe that raising awareness can start with just one person, and then spread. Recently there was a project near me to take a piece of disused land and turn it into a community garden. Just that was enough to get the attention of young people – it made us proud of our patch.

C Mina

In my city there was very little interest in raising environmental awareness until very recently so, as a result, it was hard for me to meet like-minded people, but luckily there's an active group in my area now with which I'm quite involved. And in school science classes, we're always talking about global warming, but on an everyday basis I know I've been guilty of not thinking about what I was doing to benefit the planet. I think that it should be a legal obligation to protect the environment so it becomes something we simply have to do, otherwise people just won't change, especially if it costs them money. But it's not all bad news. In local supermarkets now, you're encouraged to bring your own bags to reduce plastic waste, which is a positive move.

D Finn

In my area, it can be hard to buy stuff like organic food. Instead, like in most big cities, there are lots of companies that don't contribute positively to the environment, but you just know it'd need an impossibly massive budget to create the kind of change that would really make a difference. Solar panels and hydroelectricity are available but the government seems to be doing very little to ensure that enough use is made of them. My city's in a vulnerable location being at sea level, so these issues can't be ignored, but we've relied on oil since forever, and because of this people are unwilling to switch to alternative energy sources, even though I think that would make sense. I'm really interested in the environment but it can be hard getting others my age to share my views – that's a huge frustration for me.

WRITING (1 hour 20 minutes)

Part 1

You **must** answer this question. Write your answer in **140–190** words in an appropriate style **on the separate answer sheet**.

- 1 In your English class you have been talking about free time. Now your English teacher has asked you to write an essay for homework.

Write your essay using **all** the notes and giving reasons for your point of view.

Having a chat with friends in your free time can be as important as studying.

Do you agree?

Notes

Write about:

1. learning to get on with people
2. wasting valuable time
3. (your own idea)

Part 2

Write an answer to **one** of the questions **2–4** in this part. Write your answer in **140–190** words in an appropriate style **on the separate answer sheet**. Put the question number in the box at the top of the answer sheet.

- 2** You see a notice in an online English magazine.

Articles wanted

A Famous Guest!

If you could invite one famous person to your party, who would you choose?

Write us an article explaining why you would choose this famous person and what you would like this person to do at your party.

The best articles will be published in next month's magazine.

Write your **article**.

- 3** You see this notice in an online English magazine.

Reviews wanted

Science Fiction Films

Have you seen a science fiction film recently? Write us a review of the film giving information about the story, the actors and any special effects which were used.

Explain why you would or wouldn't recommend the film to other teenagers.

The best articles will be posted on the website.

Write your **review**.

- 4** You have seen this notice in an English magazine.

Stories wanted

Your story must **begin** with this sentence:

Nick was so excited as he sat on his bicycle, waiting for the race to begin.

Your story must include:

- an animal
- a reward

Write your **story**.

LISTENING (approximately 40 minutes, including 5 minutes' transfer time)

Part 1

You will hear people talking in eight different situations.

For questions **1–8**, choose the best answer (**A**, **B** or **C**).

- 1** You hear a science teacher telling her class about this week's lessons on the subject of food.
What is she asking them to do?

 - A** attempt a challenging new experiment
 - B** check the accuracy of their predictions about different foods
 - C** use equipment designed to examine the health benefits of cooked food

- 2** You hear a boy talking about his new basketball coach.
How does he feel about the coach's approach?

 - A** critical of his lack of attention to detail
 - B** worried about the amount of training he has organised
 - C** annoyed at the way he has treated some players

- 3** You hear a girl talking about teaching herself to draw.
She says that the best way to improve her skills was

 - A** getting guidance from professional artists.
 - B** working on things she had difficulties with.
 - C** increasing the amount of time she spent practising.

- 4** You hear a news item on the radio about an annual event at London Zoo.
What is the purpose of the event?

 - A** to attract increased numbers of visitors to the zoo
 - B** to provide a check on the animals' wellbeing
 - C** to contribute to a wider scientific research programme

- 5** You hear a boy talking about a music blog he has started.
What is he complaining about?
- A** the unhelpful response from some readers
 - B** the difficulties in trying to make it original
 - C** the pressure to add new content regularly
- 6** You hear two friends talking about a trip to a studio where a famous film was made.
What do they agree about the trip?
- A** It would even interest people who weren't fans of this type of film.
 - B** It was more enjoyable than they thought it would be.
 - C** It helped them to make sense of the book the film was based on.
- 7** You hear a history teacher talking to her students.
What is she doing?
- A** identifying the people behind a major change
 - B** warning against relying too heavily on limited information
 - C** highlighting the difficulty of finding out historical facts
- 8** You hear two friends talking about an online science quiz they did.
How does the boy feel about it?
- A** reluctant to admit he learned anything new from it
 - B** pleased that he had the opportunity to do it
 - C** surprised by his poor performance in it

Part 2

You will hear a woman called Marianne Greene answering questions about her job as a designer of theme park rides such as roller coasters. For questions 9–18, complete the sentences with a word or short phrase.

Marianne Greene – Roller Coaster Designer

Marianne says the first roller coaster that people paid to ride on used tracks which had originally

carried

9

from the mountains.

As well as subjects Marianne studied at college, she found a knowledge of

10

useful in her work.

Marianne says a design for a new ride may be adapted to suit the

11

it will be built in.

Designers are usually more concerned than theme park owners about the

12

the roller coaster will have to carry.

Marianne mentions gravity and

13

as examples of different forces

people are exposed to on roller coaster rides.

Marianne says that most of the information she uses has come from observing how

14

are affected by the job they do.

Marianne explains that most customer complaints involve rides that people think are

15

Marianne claims her latest ride will feel strange because of the type of

16

it has.

Marianne's most frightening ride includes a feature called a

17

On Marianne's favourite roller coaster, a

18

was the part she liked most.

Part 3

You will hear five short extracts in which teenagers are talking about being part of a drama group which put on a musical at school. For questions **19–23**, choose from the list (**A–H**) what impressed each speaker about the experience. Use the letters only once. There are three extra letters which you do not need to use.

- | | | |
|--|-----------|--|
| A how emotional certain people were at times | | |
| B how good some people were at problem-solving | Speaker 1 | <div style="border: 1px solid black; width: 50px; height: 20px; display: flex; align-items: center; justify-content: center;"><div style="flex: 1; background-color: #f0f0f0;"></div><div style="flex: 0.5; background-color: #d0d0d0; text-align: center; width: 20px;">19</div></div> |
| C how suitable the subject of the musical was for a school show | Speaker 2 | <div style="border: 1px solid black; width: 50px; height: 20px; display: flex; align-items: center; justify-content: center;"><div style="flex: 1; background-color: #f0f0f0;"></div><div style="flex: 0.5; background-color: #d0d0d0; text-align: center; width: 20px;">20</div></div> |
| D how much effort people in the group made | Speaker 3 | <div style="border: 1px solid black; width: 50px; height: 20px; display: flex; align-items: center; justify-content: center;"><div style="flex: 1; background-color: #f0f0f0;"></div><div style="flex: 0.5; background-color: #d0d0d0; text-align: center; width: 20px;">21</div></div> |
| E how much self-confidence some people gained | Speaker 4 | <div style="border: 1px solid black; width: 50px; height: 20px; display: flex; align-items: center; justify-content: center;"><div style="flex: 1; background-color: #f0f0f0;"></div><div style="flex: 0.5; background-color: #d0d0d0; text-align: center; width: 20px;">22</div></div> |
| F how talented some members of the group were | Speaker 5 | <div style="border: 1px solid black; width: 50px; height: 20px; display: flex; align-items: center; justify-content: center;"><div style="flex: 1; background-color: #f0f0f0;"></div><div style="flex: 0.5; background-color: #d0d0d0; text-align: center; width: 20px;">23</div></div> |
| G how well the show was received | | |
| H how much planning was required | | |

Part 4

You will hear an interview with a television presenter called Paul Murphy, who does the commentary on athletics events. For questions **24–30**, choose the best answer (**A**, **B** or **C**).

- 24** How does Paul feel about his first job as a commentator?
- A** grateful for the support he was given
 - B** proud of the way he handled it
 - C** lucky to have been offered it
- 25** Paul thinks his teaching background helped him because it meant he was used to
- A** doing a lot of preparation.
 - B** dealing with unexpected problems.
 - C** speaking to large numbers of people.
- 26** What was Paul's most memorable experience in the recent championships?
- A** speaking to spectators in the crowds
 - B** seeing several records being broken
 - C** watching the performance of a particular athlete
- 27** Paul thinks television viewers would be surprised to find out
- A** how difficult it can be to concentrate during events.
 - B** how nervous he gets before a competition.
 - C** how long his working day usually is.
- 28** What surprised Paul about one young athlete he recently interviewed?
- A** how unrealistic her ambitions were
 - B** how confident she seemed
 - C** how hard she trained

29 Paul thinks that some young athletes don't become world champions because of

- A** the lack of suitable facilities for them to train in.
- B** the number of other things they do in their lives.
- C** the problems they have recovering from injury.

30 Paul would like to commentate on competitive cycling because

- A** he's keen to understand its appeal.
- B** he wants to travel more widely.
- C** he'd like to try something different.